# Recent infection and direct genetic connectivity among recent diagnoses identify the HIV cases most likely to generate future linked diagnoses

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# Introduction

One of the crucial obstacles to achieving HIV epidemic control is the inability to reliably identify undiagnosed infected individuals. Without this knowledge, it is unclear where prevention and treatment resources will have the greatest effect. Molecular analysis offers a possible solution to this impasse. HIV-1 sequences are routinely generated for clinical drug-resistance testing but can be repurposed to reconstruct genetic networks. These networks provide a near-real-time view of ongoing transmission and help identify groups of individuals toward whom control resources can be directed. Several jurisdictions now act on this information (France et al., 2025; Poon et al., 2016); however, the best way to do so remains contested.

Prior comparisons of targeting strategies that draw on genetic networks have produced four relevant findings: (i) individual-level network-based scores prospectively predict onward transmission and, in simulation, outperform random selection for targeting (Little et al., 2014); (ii) cluster-level targeting outperforms simple individual-level targeting (Wertheim et al., 2018); (iii) machine-learning models trained on surveillance variables do not significantly improve on the CDC (the U.S. Centers for Disease Control and Prevention) priority-cluster criterion (Erly et al., 2020); however, (iv) adding network features to demographic and clinical predictors substantially improves the prediction of which individuals will be linked to future cases (Labarile et al., 2023). Two further findings bear directly on targeting: a small fraction of lineages involved in transmission bursts contribute disproportionately to subsequent infections, making them important targets (Billock et al., 2025), while in simulation, detected clusters represent only a fraction of the full transmission network, constraining the reach of any targeting strategy (France et al., 2024).

Prospective real-world implementations have yielded more nuanced lessons: using cluster information to improve contact tracing outreach was unsuccessful (Kantor et al., 2024), but integrating contact tracing and genetic networks revealed only marginal overlap between the two, indicating that each captures epidemiological links that the other misses (Khanna et al., 2025).

The UK has committed to reducing new HIV diagnoses by 90% by 2030 (Department of Health and Social Care, 2021). Although the UK Health Security Agency (UKHSA) already operates a national HIV-1 molecular surveillance pipeline, evidence on how best to use it for prevention is needed: no head-to-head comparison of candidate triage rules has been reported on the national surveillance data to which they would be applied. Here, we perform such a comparison on the UK national HIV-1 genetic network, under a design that mirrors a monthly cluster-review cadence, and identify which targeting strategy yields the most future genetically linked cases per targeted individual.

# 1 Methods

## 1.1 Data

The UK HIV Drug Resistance Database contains protease (PR) and partial reverse transcriptase (RT) gene sequences generated by the routine clinical care of people diagnosed with HIV in the United Kingdom. An automated molecular analysis pipeline (Phylotrack) detects and monitors genetic clusters prospectively, and is routinely updated with sequence, demographic, and laboratory data reported to UKHSA for surveillance. All analyses use de-identified surveillance data within a secure environment, with strict safeguards for privacy and confidentiality.

## 1.2 Genetic network

We used all PR/RT sequences from the 12 most common HIV-1 subtypes (A1, A3, A6, B, C, CRF01_AE, CRF02_AG, CRF06_cpx, CRF50_A1D, D, F1, G), comprising 91,568 individuals with PR/RT sequences sampled between 2001 and 2025. We generated a genetic network with HIV-TRACE (Kosakovsky Pond et al., 2018). Sequences were aligned to the HXB2 reference genome across the concatenated PR/RT region. Two sequences were considered genetically linked when their Tamura–Nei (TN93) genetic distance (Tamura and Nei, 1993) was at most 0.5% — the threshold used by the U.S. CDC (France et al., 2025; Oster et al., 2018). Here, a genetic cluster is any connected component of this genetic network.

An individual becomes observable in this network on the first date by which both they and at least one of their genetic partners have been sampled. We refer to this as the individual's network **entry date**. This date, rather than the sampled date alone, anchors any network-based eligibility rule defined below. To identify recently diagnosed individuals, we also use an estimated **diagnosis date**, computed as the sampled date minus the recorded months between diagnosis and sampling; when this value is missing, we fall back to the sampled date. The recency of diagnoses has been shown to predict new linked cases more strongly than their sampled dates (Chato et al., 2020).

## 1.3 Targeting rules

Each targeting rule pairs a genetic triage condition with an individual-level criterion. We use four **triage conditions** on the genetic network (Figure S1). **No genetic triage (NGT)**: any sequenced individual meeting a given individual-level criterion is eligible for targeting. **Genetic triage (GT)** is the minimal genetic network condition where any sequenced individual meeting a given individual-level criterion is eligible for targeting if there is at least one realised genetic link. **GT-cluster**, which operationalises the CDC criterion, adds a cluster size requirement on this baseline: when a cluster has accumulated at least $k$ recently diagnosed members, all qualifying cluster members (i.e., meeting a given individual-level criterion) become eligible for targeting. Finally, **GT-subnetwork** requires a connected group of recent diagnoses: when at least $k$ recently diagnosed individuals are linked to one another, all qualifying members of that group become eligible for targeting. Thus, rather than waiting for a cluster to accumulate several recent diagnoses, it acts as soon as at least two recent diagnoses are directly linked, treating a single realised recent-to-recent link as the minimal evidence of an active transmission chain.

For both GT-cluster and GT-subnetwork, individuals diagnosed before the threshold was reached but who meet the individual-level criterion become eligible for targeting at the moment the threshold is first reached.

For any GT rule, the targeted set can be further extended by including each targeted individual's direct genetic neighbours sampled at the time they were first targeted — regardless of whether those neighbours themselves meet the individual-level criterion — while respecting the budget cap of $N$. We refer to this as the neighbour extension, mirroring operational practice, in which the diagnosis of an index case prompts elicitation and follow-up of named partners.

We use four individual-level criteria. Three of them — recent diagnosis, recent infection, and high viral load — rely solely on the individual's record. The fourth, high degree, is computed from the network itself and is therefore paired only with triage conditions that also use the network.

**Recent diagnosis** is the default criterion and defined as an estimated diagnosis date within the preceding 12 months. The remaining three add a further requirement on it: **recent infection** requires a positive Recent Infection Testing Algorithm (RITA) test result; **high viral load** requires a viral load above $10^5$ copies/mL at sampling; **high degree** requires a network degree of at least three at the individual's network entry date — that is, at least three of their genetic partners had themselves been sequenced by the time the individual first became observable in the network.

## 1.4 Evaluation framework

To compare targeting rules under conditions that mirror operational surveillance, we applied each rule monthly over 8 years (December 2011 — December 2019), so that performance is assessed across varying epidemiological and data-availability conditions. In each month, eligible individuals are targeted in order of arrival — by sampled date for NGT rules, and by network entry date for GT, GT-cluster, and GT-subnetwork rules (reflecting when an individual first becomes observable in the network) — until the budget of $N$ individuals is filled. We set $N = 250$, a round figure on the scale of one month's national HIV diagnoses in England (2,773 in 2024 and 2,838 in 2023, $\sim$230–240 per month (UK Health Security Agency, 2025)), and held it constant across the evaluation window so that every rule is subject to the same budget ceiling. Outcomes are then measured from the date each individual was first targeted, over a one-year forward window.

## 1.5 Outcome measurement of targeted individuals

The outcome at each month is the per-target yield: the number of newly diagnosed, genetically linked cases attributed to the targeted set during the forward window, divided by the size of the targeted set. Because targeting under GT-cluster and GT-subnetwork cannot begin until the threshold is reached, neither rule is credited for outcomes that pre-date its first opportunity to act.

Attributing a new case to a specific targeted individual is not straightforward because a newly diagnosed case may be linked to multiple individuals already in the network, and genetic data alone do not identify which of them, if any, transmitted to the new case (Wertheim et al., 2018). We therefore weight each new case by the inverse of its degree at the time of diagnosis — the number of genetic partners already sampled by then — thereby distributing credit evenly across the potential sources. A targeted individual linked to the new case is credited with this fraction. For example, if a new case is linked to four individuals already in the network and one of them is in the targeted set, that target is credited with $1/4 = 0.25$ of a case; if two of the four are in the targeted set, each is credited with 0.25, and the new case contributes 0.5 in total. Formally, the contribution of target $i$ to new case $Y$ is

$$c(i,Y) = \begin{cases} 1/d(Y) & \text{if } i \text{ and } Y \text{ are genetically linked and } Y\text{'s} \\ & \text{estimated diagnosis date falls in } i\text{'s forward window,} \\ 0 & \text{otherwise,} \end{cases}$$

where $d(Y) = |\{Z : (Y,Z) \in E,\ \text{sampled}(Z) < \text{sampled}(Y)\}|$ is the number of $Y$'s genetic partners already sampled before $Y$, and $E$ is the edge set. The yield reported each month is $\frac{1}{|T|}\sum_i \sum_Y c(i,Y)$, where $T$ is the set of targeted individuals in that month.

### 1.6 Software and reproducibility

All analyses were implemented in R and are available at [TBA].

## 2 Results

We compared 15 targeting rules, comprising a genetic triage condition (none, NGT; a single realised link, GT; or a cluster- or subnetwork-size gate, GT-cluster and GT-subnetwork; Figure S1) and four individual-level criteria (recent diagnosis, recent infection, high viral load, high degree). For each rule, we report the per-target yield — newly diagnosed, genetically linked cases attributed per targeted individual over a 12-month forward window — as a difference from the operational baseline (NGT × recent diagnosis: recently diagnosed sequenced individuals targeted in order of arrival, with no genetic triage), across 97 monthly evaluations.

We first characterised the genetic network as a whole. Of the 91,568 individuals sequenced between 2001 and 2025, roughly one-third (33,614; 36.7%) were genetically linked to at least one other at the 0.5% threshold, forming a network of 4,975 clusters and 49,769 edges (Figure S2).

We then characterised the pool each targeting rule could draw on over the evaluation window (December 2011–December 2019), comprising 33,033 recently diagnosed individuals across the 12 most common subtypes. This number closely tracked national HIV diagnoses in England (mean 86.6% of reported diagnoses). An estimated diagnosis date was available for 24,816 of them (75.1%); among these, sampling coincided with diagnosis for 81.9%, and for the remainder diagnosis preceded sampling by a median of 3 months (IQR 1–11). Individuals were predominantly men who have sex with men (MSM, 42.5%) or people reporting heterosexual exposure (31.3%), with people who inject drugs and other exposures together under 5%; exposure was unrecorded for 22.2% of cases (Table S1). MSM tested positive for RITA far more often (16.2% of the group) than any other exposure group (all $\leq$7%), whereas the proportion with high viral load varied little among the main groups ($\sim$20–22%).

Across the 97 monthly evaluations, both the full sequenced pool and its in-network subset filled the monthly budget of $N = 250$ recently diagnosed individuals in every month (minimum of 1,720 and 639 individuals available for targeting, respectively) (Figure S3). However, the individual-level criteria differed in the number of individuals available for targeting per month. In the sequenced pool, high viral load reached the budget in 95% of months (mean 599 eligible per month), and recent infection (RITA+) reached the budget in 74% of months (mean 330 eligible). Once restricted to the in-network pool, high viral load and high degree reached the budget in 64% and 68% of the 97 months, respectively (mean 263 and 317 eligible), and RITA+ in only 29% of months (mean 186 eligible).

The operational baseline (NGT $\times$ recent diagnosis) yielded a median of 0.109 attributed linked cases per targeted individual across the 97 monthly evaluations (IQR 0.087–0.134). Without any genetic triage, an individual-level biomarker raised yield over the baseline: recent infection by +0.097 and high viral load by +0.025 (Figure 1). Adding a genetic triage on top of an individual-level criterion raised per-target yield beyond the individual-level criterion alone; whether it also increased the total number of cases found per month varied by criterion, as we show below. The largest yields came from high degree — a network-derived individual-level criterion — paired with GT-cluster or GT-subnetwork (+0.257 and +0.311, respectively).

We extended the original targeted set to include the direct genetic neighbours of targeted individuals—mirroring the partner follow-up that an index diagnosis could prompt in practice (Figure S4). The neighbour extension raised yield for all but high-degree targeting, where each target's direct neighbours are largely individuals the rule would otherwise select; the extension therefore diluted rather than broadened the targeted set. The largest gain was in GT-cluster × recent infection (+0.317), roughly matching GT-subnetwork × high degree without the extension (+0.311).

We decomposed each rule's yield — the number of newly diagnosed, genetically linked cases attributed per targeted individual — by the RITA status of those attributed cases at the time of their own diagnosis (Figure S5). Across all 15 rules, the decomposition was similar: on average, 22.0% of attributed cases were RITA-positive at sampling (range across rules 16.0–27.8%), 30.1% were RITA-negative (25.1–34.4%), and 48.0% had no RITA result available (40.8–55.9%).

Per-target yield captures efficiency but not absolute impact, as rules that do not exhaust the budget act on fewer individuals. We therefore also examined potential reach — the number of attributed linked cases per month (yield × number targeted; Figure S6). Targeting recent diagnoses in order of arrival until the budget cap (the operational baseline) reached a median of 27 cases per month. High-degree targeting achieved the greatest reach under every genetic triage condition (medians 68–81), reflecting both a high per-target yield and an eligible pool large enough to approach the budget. Recent infection (RITA+) also raised reach substantially on its own (median 52 under NGT); here, however, genetic triage was counterproductive, increasing per-target yield but reducing reach (to 22–31 under GT-cluster and GT-subnetwork), because the joint requirement of RITA-positivity and clustering left too few eligible individuals to target (median 62–141 individuals targeted).

In a sensitivity analysis reducing the forward observation window to 6 months, the ranking of targeting rules remains invariant — high-degree targeting under genetic triage remained the top performer and recent infection (RITA+) the leading single-feature criterion — and all gains over baseline persisted at roughly two-thirds of their 12-month magnitude.

RITA+ is still the best single-feature criterion, but only barely. (selecting RITA+ in the all-data run is implicitly selecting "MSM in active transmission"–figure S7)

**BUT**, when it comes to 'reach', the high-degree eligible pool is reduced, high-degree reach roughly halves ( 25–39) and no longer dominates

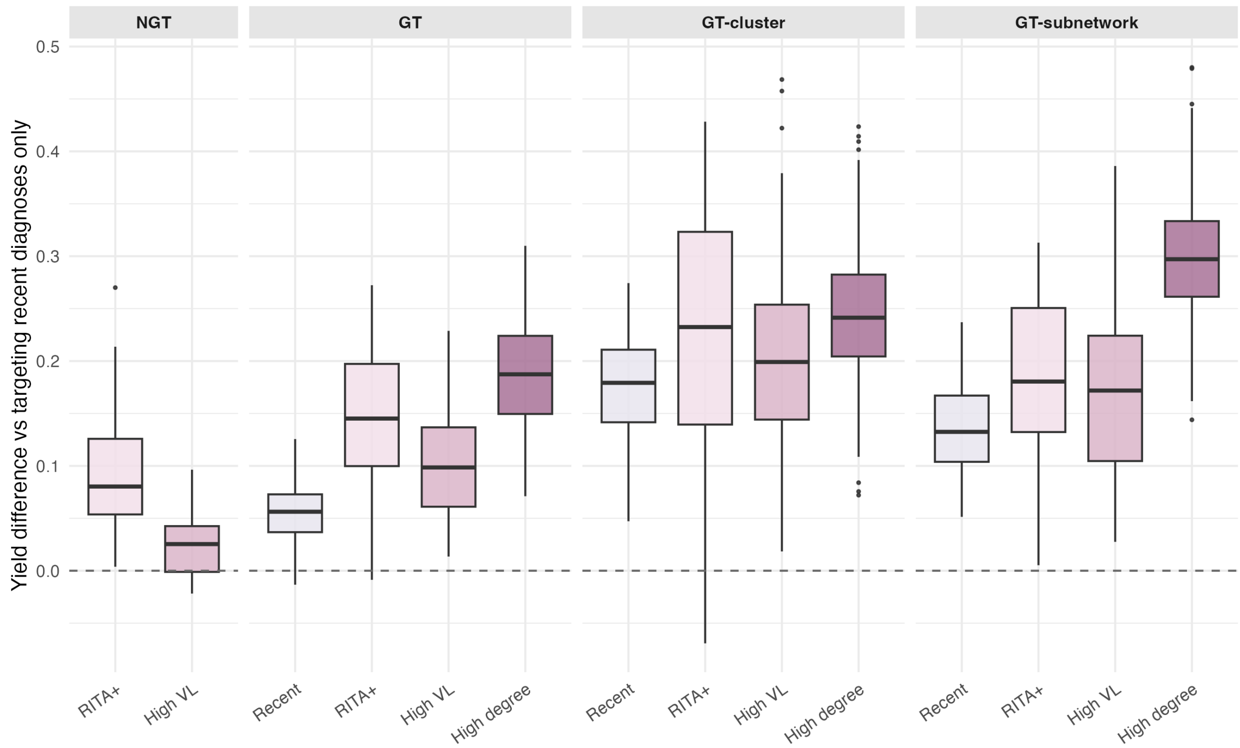


Figure 1: **Individual-level biomarkers improve yield over the operational baseline of targeting any recent diagnosis; genetic triage adds further improvement, with the largest gains coming from high-degree targeting.** Per-month yield differences across 97 monthly evaluations (December 2011 — December 2019) with a budget cap of $N = 250$. Yield is the edge-weighted attribution of newly diagnosed linked cases over a 12-month forward window. Baseline: NGT $\times$ recent diagnosis (recently diagnosed sequenced individuals targeted in order of arrival, with no genetic triage).

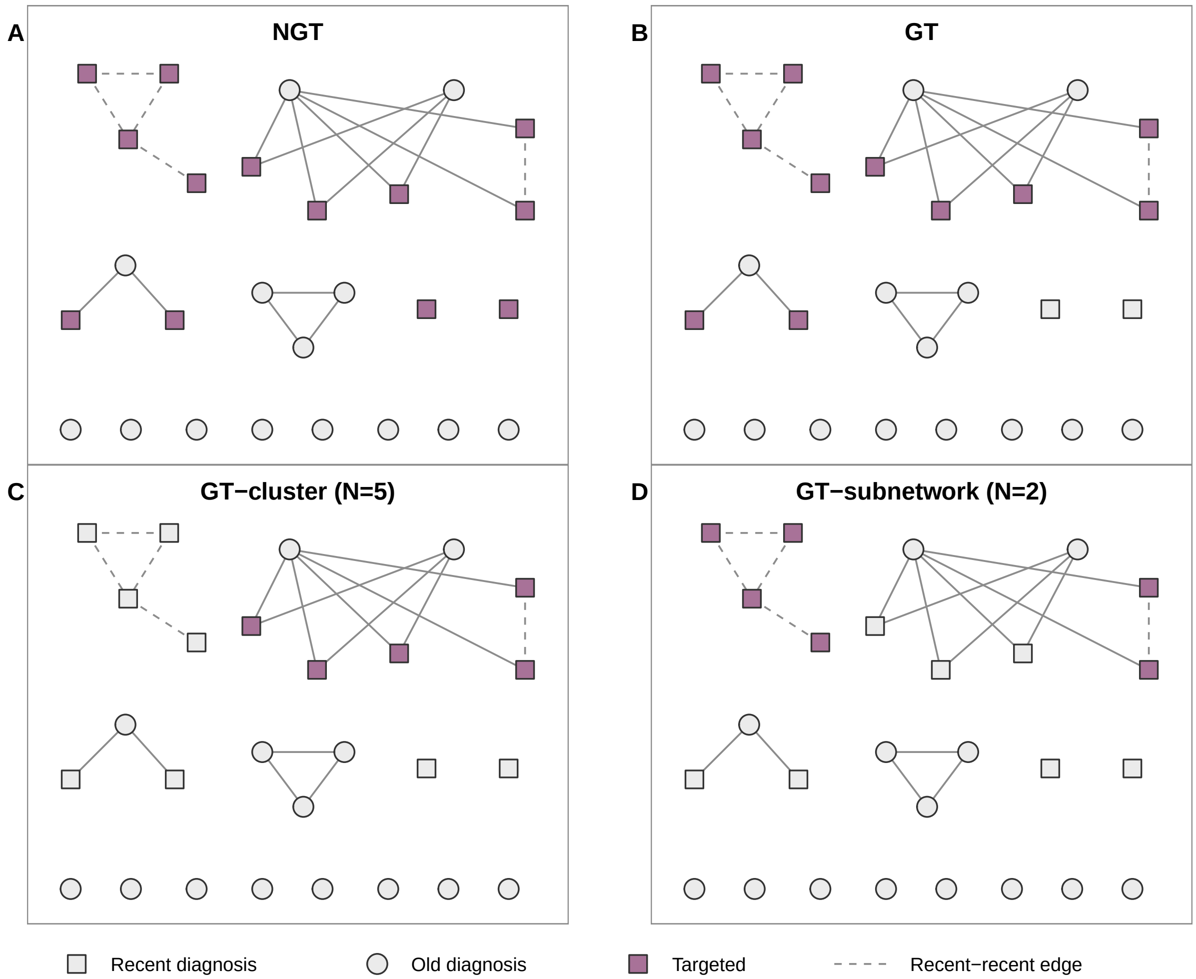


Figure S1: **Schematic of the four triage conditions on the genetic network.** Each panel applies one rule to the same illustrative network: squares represent recent diagnoses (within the preceding 12 months), circles represent older diagnoses, orange-filled nodes mark individuals targeted by the rule, and dashed lines link two recent diagnoses. (A) NGT targets every recent individual regardless of network membership. (B) GT also requires at least one realised genetic link. (C) GT-cluster ($k = 5$) restricts targeting to individuals whose cluster contains at least 5 recent members (the CDC priority-cluster operationalisation). (D) GT-subnetwork ($k = 2$) restricts targeting to individuals in a recent-to-recent connected component of size at least 2. The illustration uses recent diagnosis as the individual-level criterion; the same logic applies to RITA+, high viral load, or high degree.

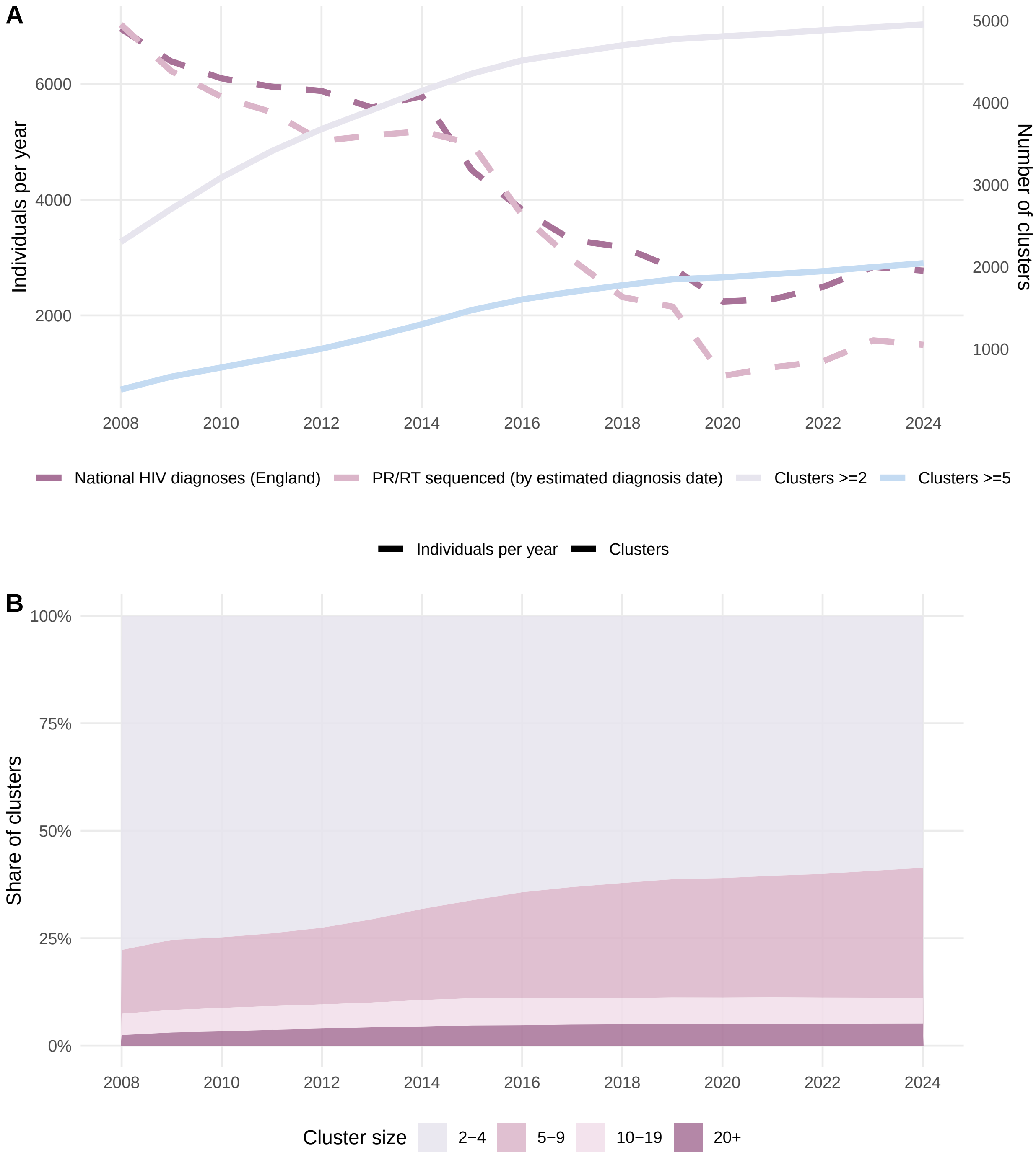


Figure S2: **PR/RT-sequenced individuals, national HIV diagnoses, and the number of genetic clusters at 0.5% TN93 distance.** (A) Annual sequenced individuals (blue, by estimated diagnosis date) versus national diagnoses (red) on the left axis; cumulative end-of-year cluster counts ($\geq 2$ and $\geq 5$ members) on the right axis. (B) End-of-year cluster size distribution as the share of clusters in each size bin.

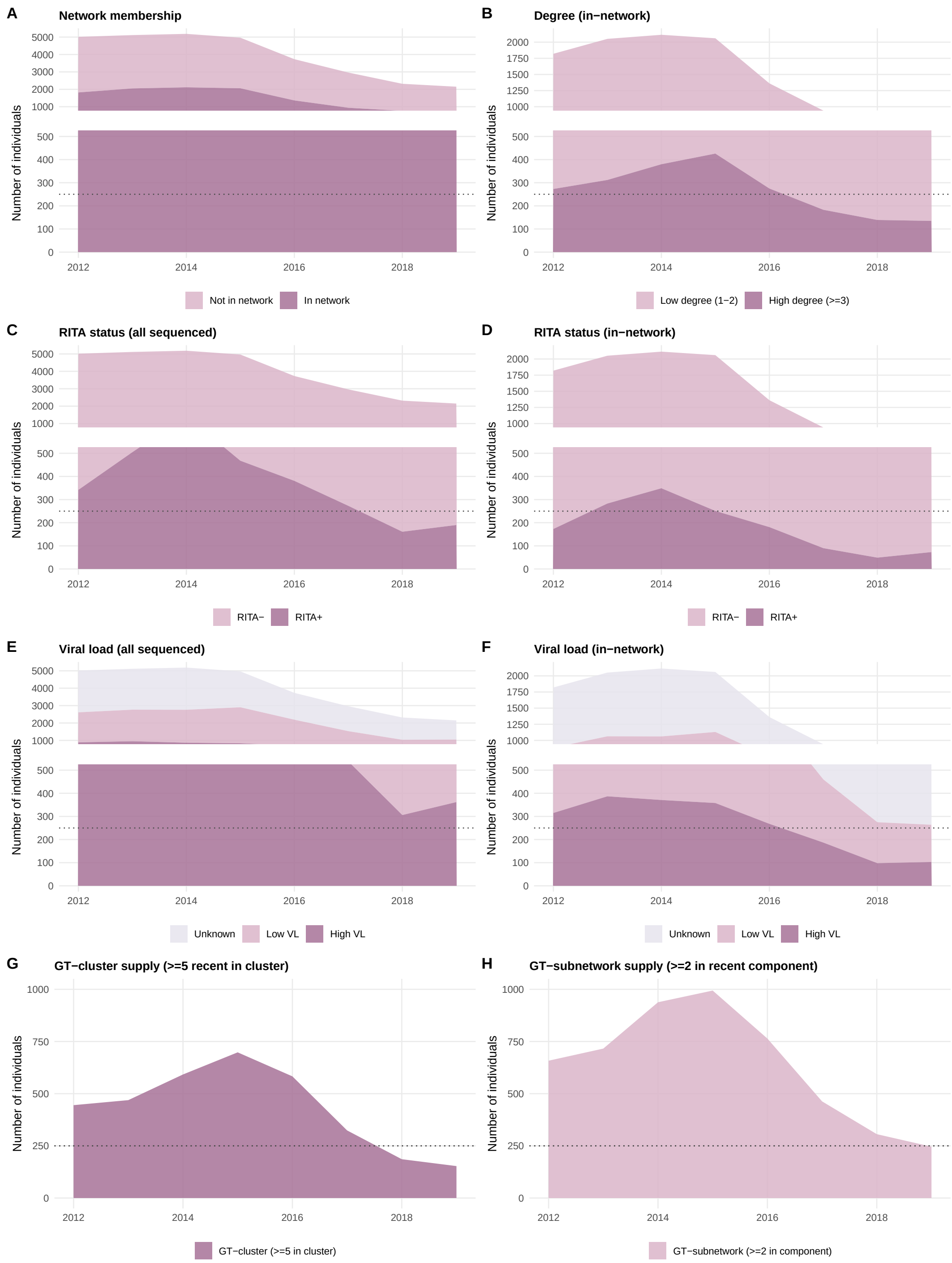


Figure S3: **Availability of individual-level criteria across the sequenced and in-network populations, and the monthly supply under each genetic triage condition.** (A–F) Annual breakdown of the PR/RT-sequenced pool by network membership, degree at network entry, RITA status (recent infection at sampling), and viral load (dichotomised at $10^5$ copies/mL), shown separately for the full cohort and the in-network subset. (G, H) Monthly count of individuals eligible under GT-cluster ($\geq 5$ recent members in the same cluster) and GT-subnetwork ($\geq 2$ recent in a connected component), averaged within year. Dotted line in A–H: targeting budget $N = 250$.

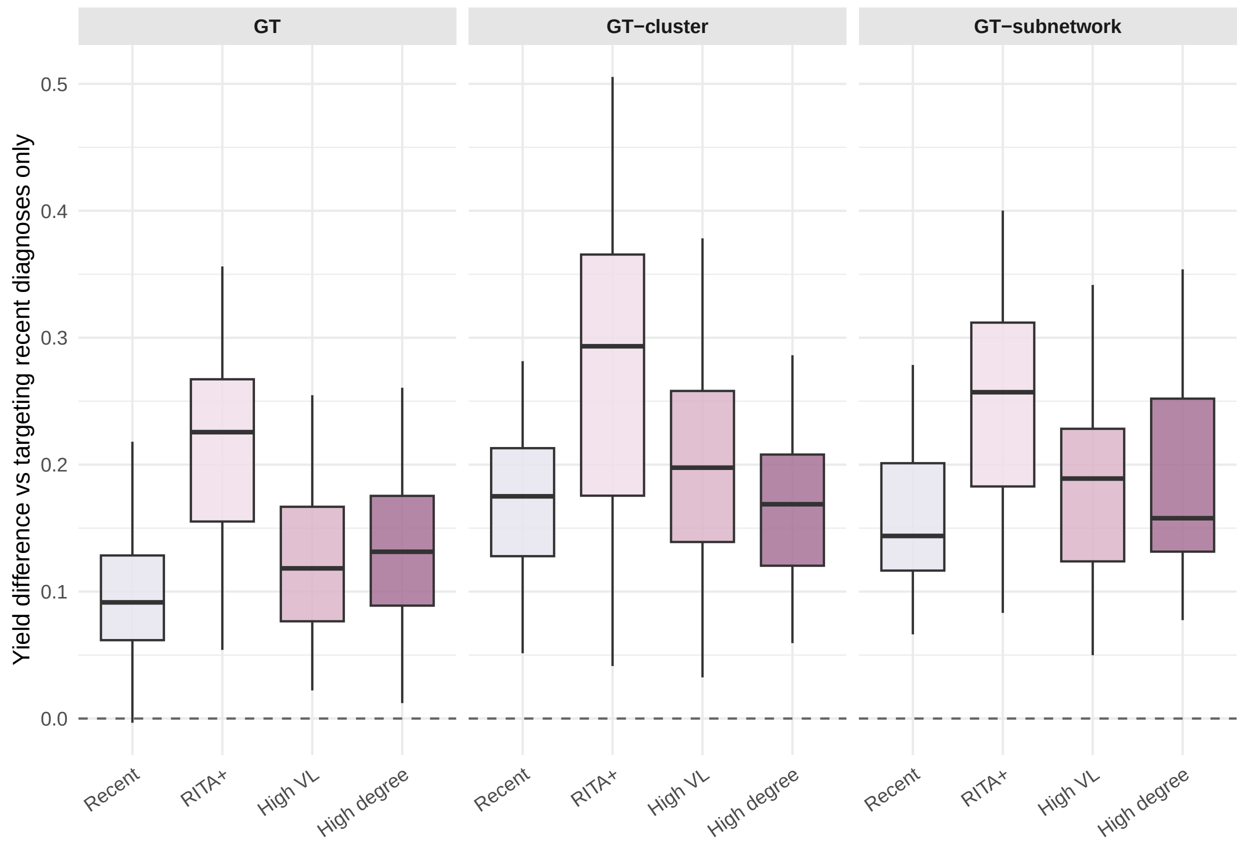


Figure S4: **The neighbour extension raises yield for every rule except high-degree targeting.** Per-month yield differences across 97 monthly evaluations (December 2011 — December 2019). Each targeted set is extended to include the direct genetic neighbours of the targeted individuals, within the budget cap of $N = 250$ targeted individuals per evaluation. Yield is the edge-weighted attribution of newly diagnosed, genetically linked cases over a 12-month forward window. Baseline: NGT $\times$ recent diagnosis (recently diagnosed sequenced individuals targeted in order of arrival, with no genetic triage; no neighbour extension applied). NGT is omitted from the panels because the neighbour extension is undefined without a genetic triage.

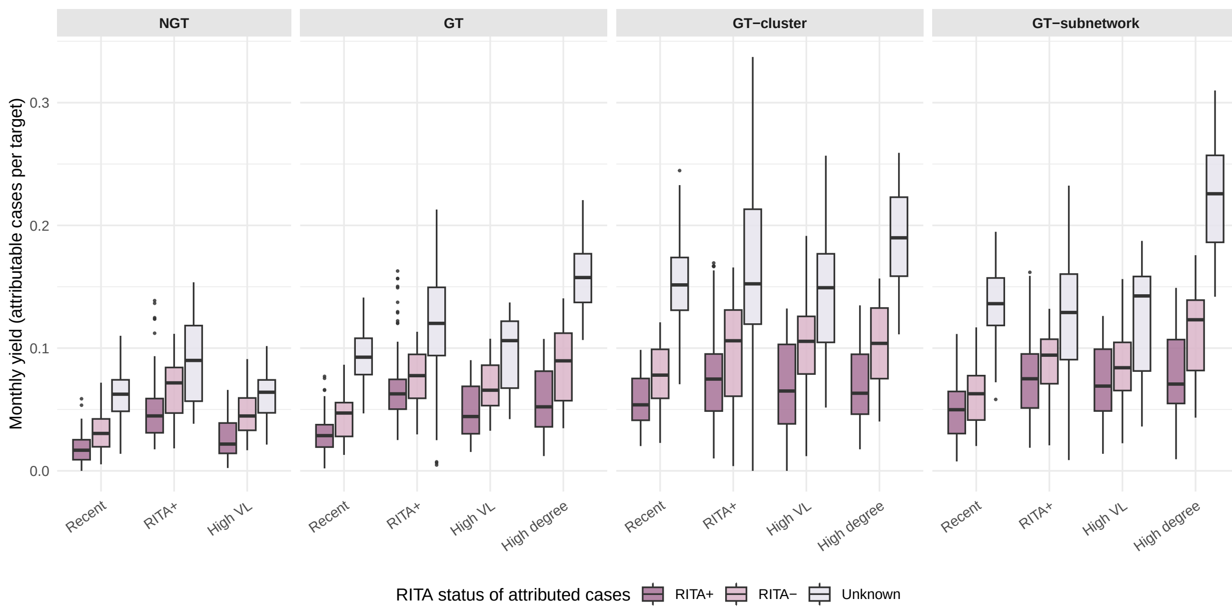


Figure S5: **The composition of per-target yield by RITA status is largely invariant across targeting rules.** For each of the 15 targeting rules, the monthly per-target yield (newly diagnosed, genetically linked cases per targeted individual) is split into three sub-yields by the Recent Infection Testing Algorithm (RITA) status of the attributed case at the time of its own diagnosis: RITA-positive (recent infection at sampling), RITA-negative (not recent), and no RITA result available. Each box summarises 97 monthly evaluations (December 2011 — December 2019) with a budget of $N = 250$ per evaluation.

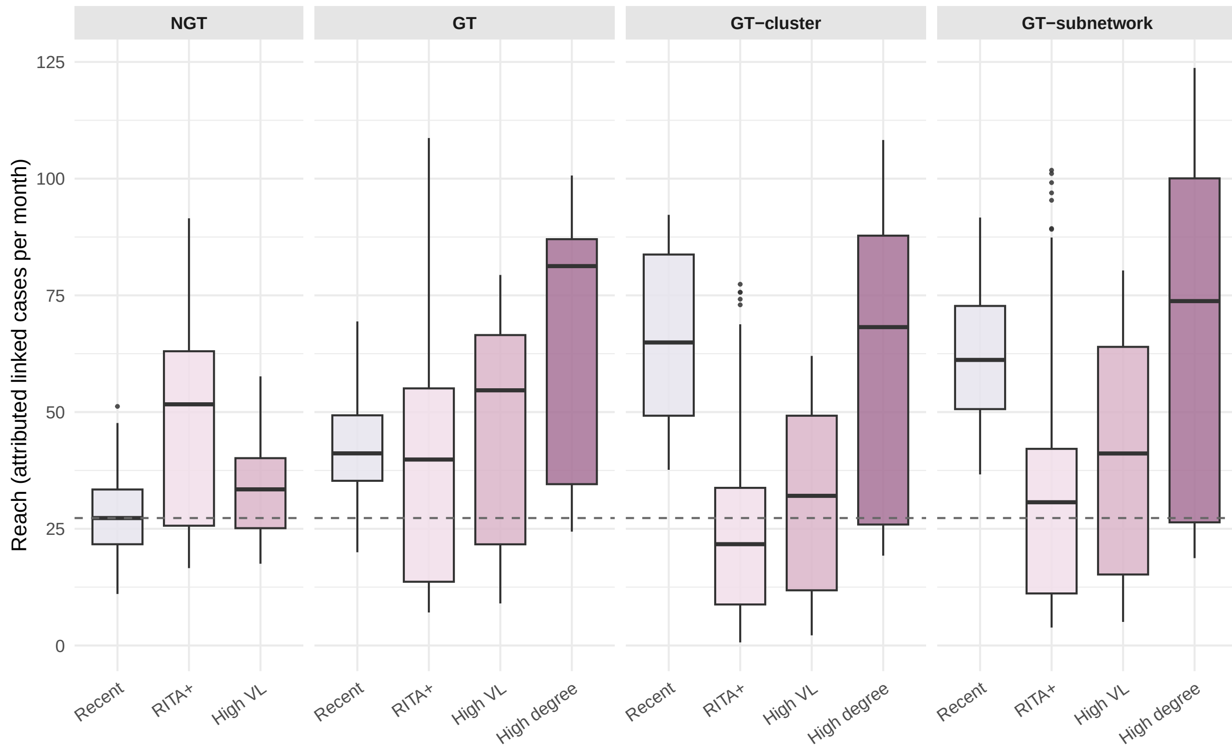


Figure S6: **High-degree targeting achieves the greatest absolute reach, whereas recent infection reaches most when applied on its own**. Per-month absolute reach — the number of newly diagnosed, genetically linked cases attributed to the targeted set (yield $\times$ number targeted) — across 97 monthly evaluations (December 2011 — December 2019) with a budget cap of $N = 250$. Yield is the edge-weighted attribution over a 12-month forward window. Each box summarises the 97 monthly values, grouped by genetic triage condition (NGT, GT, GT-cluster, GT-subnetwork) and individual-level criterion. Dashed line: median reach of the operational baseline (NGT $\times$ recent diagnosis: recently diagnosed sequenced individuals targeted in order of arrival, with no genetic triage).

Table S1: **Exposure-group composition and biomarker ascertainment of the recently-diagnosed evaluation pool (December 2011–December 2019.** Each cell shows the count and its percentage of the relevant subset. RITA+ is a likely recent infection at sampling, and a high viral load is $> 10^5$ copies/mL.

| | | RITA | | Viral load | |
|---|---|---|---|---|---|
| Exposure group | $N$ (%) | $n$ (%) | RITA+ (%) | $n$ (%) | High (%) |
| MSM | 14,053 (42.5) | 7,830 (55.7) | 2,281 (16.2) | 9,198 (65.5) | 3,111 (22.1) |
| Heterosexual | 10,347 (31.3) | 5,249 (50.7) | 538 (5.2) | 6,971 (67.4) | 2,238 (21.6) |
| PWID | 696 (2.1) | 222 (31.9) | 43 (6.2) | 430 (61.8) | 139 (20.0) |
| Vertical | 446 (1.4) | 84 (18.8) | 7 (1.6) | 294 (65.9) | 37 (8.3) |
| Other | 163 (0.5) | 40 (24.5) | 6 (3.7) | 104 (63.8) | 28 (17.2) |
| Unknown | 7,328 (22.2) | 459 (6.3) | 49 (0.7) | 598 (8.2) | 145 (2.0) |
| Total | 33,033 (100) | 13,884 (42.0) | 2,924 (8.9) | 17,595 (53.3) | 5,698 (17.2) |

# 3 Discussion

Using UK national HIV-1 molecular surveillance data, we compared 15 targeting rules across 97 monthly evaluations under a design that mirrors a monthly cluster-review cadence and identified which delivered the most genetically linked future diagnoses per targeted individual. The largest gains over an operational baseline (targeting any recent diagnosis) came from combining a network-derived individual-level criterion with a genetic triage gate: targeting high-degree individuals within clusters or within recent-to-recent genetic links — more than tripling the baseline yield of 0.109. Among the criteria that rely only on individual-level data, recent infection (RITA+) was the dominant single-feature driver. By quantifying gains prospectively on a national surveillance stream, these results extend prior comparisons that established cluster-level targeting as superior to simple individual-level targeting (Wertheim et al., 2018) and that network features improve the prediction of which individuals will be linked to future cases (Labarile et al., 2023).

Individual-level information improved yield, and genetic triage amplified these gains. Coupled with the most selective criteria, however, genetic triage also acted on fewer individuals, since these rules did not always fill the monthly budget, and whether this raised or lowered absolute reach depended on the criterion. For high degree, the eligible pool stayed large enough that the higher per-target yield outweighed the smaller number targeted. For recent infection, by contrast, the joint requirement of RITA-positivity and clustering left too few eligible individuals, and reach fell to around or below baseline; RITA+ delivered its greatest reach when applied on its own. These patterns suggest a tiered strategy: recent infections can be acted on immediately through routine testing, without waiting for cluster membership to be established, whereas, where network data are available, the greatest impact comes from prioritising recently diagnosed, well-connected individuals within clusters.

Realising any of these rules in routine national surveillance, however, depends not only on laboratory throughput but also on promptly linking sequence data to clinical and demographic records for action. In practice, this linkage is often the binding constraint: integrating traditionally siloed genetic, epidemiological, and clinical data remains difficult even in mature response programmes (Oster et al., 2021), and without timely access to it, a real-time targeted strategy is hard to mount (Obeng et al., 2024). This frames a policy choice: tightening cross-system data integration would help unlock the higher per-target yield of these rules at programmatic scale, supporting progress toward HIV elimination goals, whereas GT-cluster × recent diagnosis (the CDC operationalisation) requires no additional information and remains a pragmatic option.

The two criteria with the largest single-feature gains over the baseline — RITA-positive and high degree — both likely identify individuals embedded in currently active transmission, by different routes. RITA+ signals individuals with recent infection, and thus a window of elevated viral load and per-contact transmission probability (Bellan et al., 2015); such recently infected individuals are themselves more likely to be found within actively growing clusters (Di Giallonardo et al., 2021). Consistent with this, the largest gain from the neighbour extension came from GT-cluster × recent infection (+0.317). For high degree, the criterion identifies the key individuals in transmission hotspots, which is why the neighbour extension increased yield for every other rule but this one. Because high degree is computed from the network itself, it is observable only where genetic data exist and is therefore a direct benefit of molecular surveillance. Convergent evidence comes from social and spatial network studies in a longitudinal cohort of people who inject drugs, where individuals who seroconverted had higher degree centrality (and acquisition risk rose with each additional viraemic network partner) (Clipman et al., 2022).

The public-health benefit of the per-target yield can be partially interpreted by the RITA status of each newly diagnosed linked case: for RITA-negative cases, the benefit is limited to earlier diagnosis (individuals were infected before the targeted individual was sampled, so targeting could not have prevented their infection), whereas for RITA-positive cases, the benefit may also include preventing onward transmission. However, most cases lacked a RITA result, limiting confident quantification of this mix. Even so, the composition was largely invariant across rules, suggesting that targeting rules differ in the number of attributed cases per target rather than in the kinds of benefit those cases represent.

Several considerations bound the interpretation of these results. First, the per-target yield is an upper bound on the public-health benefit of targeting — whether earlier diagnosis or averted onward transmission— because it credits the targeted individual with every attributed linked case as if intervention fully delivered that benefit, whereas in practice not every selected individual can be reached and interventions are only partially effective. Second, the direction of transmission is unobserved; the attribution weights each new case by the inverse of its sampled-partner degree at the new case's diagnosis, using the sampled date as a proxy for temporal order, but a partner sampled before the new case may in fact have been infected later. Third, the genetic network captures only links at TN93 distance $\leq 0.5\%$ between sequenced individuals and only $\sim$90% of UK diagnoses were sequenced at the time of this analysis, with undiagnosed cases absent altogether, so the pool any targeting rule could act on is likely larger than these yields imply. This gap has been quantified elsewhere: complete clusters were 3–9 times larger than their detected counterparts, with around a third of members undiagnosed (France et al., 2024). Fourth, the high-degree criterion is sensitive to sampling effort, and future links attributed to them may reflect dense local sampling rather than genuine transmission centrality (Chato et al., 2020).

Importantly, molecular surveillance and network-guided targeting raise ethical and equity considerations that bound how these or similar rules should be deployed. Because cluster membership depends on who has been sequenced, individuals can be flagged for action based on data generated for clinical, not surveillance, purposes, with implications for consent and confidentiality. We underscore that our high per-target yield indicates only that a rule is potentially epidemiologically effective, not that it is appropriate to deploy; any operational use should be paired with explicit governance, community engagement, and safeguards proportionate to the populations most affected (Schuster et al., 2025).

In summary, under the assumption that genetic links are a proxy for transmission, our results point to a tiered prioritisation. Recent infection (RITA+) identifies high-yield individuals from routine testing alone and is best acted on directly, without waiting for them to accrue cluster membership. Where genetic network data are available, the largest impact comes from prioritising recently diagnosed individuals who are highly connected within clusters of recent diagnoses. As the UK pursues its 2030 elimination commitment, we provide a reference for planning prevention-focused evaluations of molecular targeting strategies.

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